

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. 8494

AY: 2025-26

Sem: 1

Exam Type: Written

Session: Odd MSE October 2025

Slot: 10:00am - 11:00am

Block: A01

Course Section: 1

Date: 08-10-2025

C. Code: CS413

Seat No: 2023601001

L 29_525_986_1_1505_38339

Name of the candidate. Asar Naufel Abdul Nasir

Candidate's UID number: 2023601001

Examination class room number: 401

Invigilator's signature with date: [Signature] 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH MSE

(For example FY MCA/FY BTECH)

SEMESTER: 7

PROGRAM: CSE-AIML

DAY AND DATE: 8/10/25 Wednesday

COURSE NAME: Deep Learning

~~I~~ / ~~II~~ / ~~III~~ / ~~IV~~ / ~~V~~ / ~~VI~~ / VII / ~~VIII~~

(To be filled in by the examiner)								
Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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Question
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Q.1 a

↳ Given:

$$f(w) = \frac{1}{2} \|Xw - y\|^2$$

Gradient Descent:

$$f_{t+1}(w) = f_t(w) - \eta \nabla f_w(w)$$

where,

η is the learning rate,

$\nabla f_w(w)$ is the first order
derivative of $f(w)$
w.r.t. parameter w .

~~$$f_w(w) = \|Xw - y\|$$~~

$$\nabla f_w(w) = \frac{1}{2} \times 2 \times \frac{\partial}{\partial w} (Xw - y)$$

$$= \|X\|$$

Gradient for $f_w(w) = [X]$.

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For convergence,
we require η such that

$$0 < \eta < \frac{2}{\lambda_{\max}}$$

where $\lambda_{\max}$ is the highest value
of eigenvalues of the Hessian
matrix of $f_w(w)$ w.r.t w . ^{Jacobian}

This is due to the fact that
eigenvalue decomposition of X
gives out the equivalent
of eigenvalues of covariance
of $X^T X$. These in turn
specify the most important i.e.
highest variation planes in
eigenvectors which is the
same as reconstruction loss
 L_{rec} of the gradient update
rule which relies on

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Jacobian matrix of $f_w(w)$.

For a $f(x) = 3x^2 + 4x + 2$

$\nabla f(x) = 6x + 4$

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Q.2 a

↳ The classic convolutional neural network applying 2D convolution for $H \times W \times C$ with K filters of $F \times F$, at each layer,

$$O_H = \left\lfloor \frac{H + 2P_H - F}{S} \right\rfloor + 1$$

where the output ~~in~~ dimensions of the layer will be:

$$\Rightarrow O_H \times O_W \times \max(C, K)$$

where O_W is Output width with the same formula.

Now, we know that the Convolution operation can be represented as matrix multiplication by using Toeplitz matrix.

$$H = \begin{bmatrix} h_1 & 0 & 0 & 0 \\ h_2 & h_1 & 0 & 0 \\ 0 & h_2 & h_1 & 0 \\ 0 & 0 & h_2 & h_1 \end{bmatrix}$$

Where H is a Kernel for convolution.

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Thus, multiplying matrix by
Strassen's algorithm,

a	b
c	d

e	f
g	h

$$\Rightarrow ae + bg + ce + df$$

so in total 7 multiplications.

Therefore,

for one layer,

$$O(D_H \times D_W \times C + F)$$

$$= O(D_H \cdot D_W \cdot C)$$

We use im2col to make
matrix representation more
easier, thus.

$$\Rightarrow O(L \cdot O^2)$$

↑
output dimensions for
square matrix

Depthwise separable convolutions

are important due to

Universal Approximation Theorem,

which states that,

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a single hidden layer with any finite no. of neurons can approximate any & linear function.

Thus, having separate layers work on separate features help Deep Neural nets to identify complex patterns.

Similarly, using a FC layer directly on Input would require very high no. of neurons to generalize, so Convolution layers use less weights by convolving and pooling to increase efficiency.

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Q.2 b.

↳ Linear undercomplete autoencoder is represented by:

$$h(x) = Wx$$

and

$$g(x) = V \cdot h(x) = VWx$$

where the no. of hidden neurons (h) is lesser than input parameters x .

This architecture forces neural net to learn patterns and create an encoded representation of input while minimizing loss for reconstruction, L_{rec} .

PCA is used to find most important features, by using

eigenvectors of covariances of

XX^T such that input is

reconstructed with minimal loss.

It encodes the input embedding subspace into latent dimensions by reconstruction of covariance.



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An overcomplete encoder on other hand can learn the identity because the network is guided to reduce the reconstruction loss.

to reduce it, the network with more or equal no. of neurons to input will learn to generate the same input sequence with param weights = 1 for $h=n$.

Thus, we use constraints over AE such as:

- Sparse AE
- Denoising AE
- Contractive AE

because an encoded representation of size $\geq$ input is of no use to us.

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: 0.3a

↳

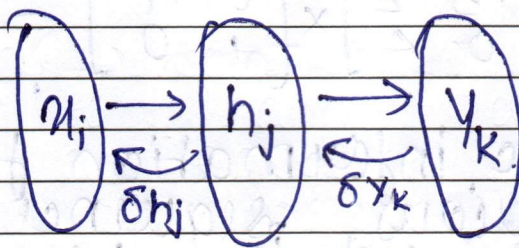
For RNN,

$$h_t = f(\cancel{h_{t-1}} w_h)$$

$$h_t = f(w[h_{t-1}, x_t] + b_h)$$

$$y_t = f(w_y \cdot h_t + b_y)$$

The gradient of RNN is a partial differentiation, which we solve using chain rule, due to backpropagation



$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial h_{t-1}} \times \frac{\partial h_{t-1}}{\partial h_{t-2}} \times \frac{\partial h_{t-2}}{\partial h_{t-3}} \times \dots \times \frac{\partial h_0}{\partial h_{t-k}}$$

when propagation occurs from a very deep (large sequence) the gradient reaches close to 0, thus relevant context vanishes with the gradient.

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Forget gate in LSTM:

$$f_t = f(W[h_{t-1}, x_t] + b_f)$$

$$\tilde{C}_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

where $*$ represents element wise multiplication.

This actually does a matrix multiplication as f_t contains 0 and 1s.

So with such formulation

$$\begin{bmatrix} 1 & 5 \\ 0 & 2 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$

specific information from previous sequence is removed to keep only relevant context.

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Q.1 b.

↳

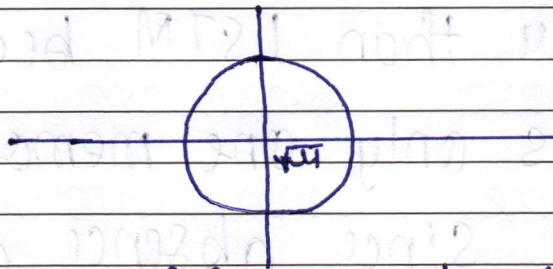
Bias and variance decomposition of an expected square error predictor is a convex quadratic loss, i.e.

$$MSE = \|w\|^2 = \frac{1}{N} \sum (y - \hat{y})^2$$

where $\|w\|^2$ represents

L_2 norm,

this restricts the loss into a fixed boundary and penalizing others.



where it restricts all values of the function $J(\theta)$ $\theta < \sqrt{M}$.

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Q.3 b.

↳ Weights in LSTM:

$$f_t = f(w_f[h_{t-1}, x_t] + b_f)$$

$$i_t = f(w_i[h_{t-1}, x_t] + b_i)$$

~~$$c_t = f_t * c_t$$~~

$$\tilde{c}_t = f(w_c[h_{t-1}, x_t] + b_c)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{c}_t$$

$$o_t = f(w_o[h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(c_t)$$

The gradient updation of GRU

uses gates to retain info,

lesser than LSTM because it

uses only one memory cell

h_t . Since absence of C_t

~~$$u_t = u_t * +$$~~

memory cell, only h_t is

applied gates (update and reset)

∴ GRU has lesser parameters.

[illegible]

